Deep Learning for Stock Price Prediction: Transformer vs LSTM with Multi-Source Sentiment Analysis

CS 7643 - Deep Learning

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Abstract

Stock price prediction remains challenging due to market noise and non-stationarity. We investigate whether deep learning models enhanced with multi-source sentiment analysis can improve next-day price direction prediction. We implement and train LSTM and Transformer architectures on four tech stocks (AAPL, NVDA, TSLA, AMZN) using technical indicators combined with StockTwits and NewsAPI sentiment (2020-2022 training, 2026 validation). Key findings: (1) Transformer outperforms LSTM by +4.12% (56.39% vs 52.27%), with NVDA achieving 62.26% accuracy; (2) combining sentiment sources improves volatile stocks (TSLA +6.08%); (3) 2026 validation shows -3.56% degradation with ticker-specific behavior; (4) multi-class LSTM suffers severe overfitting (92% train $\rightarrow$ 33% test); (5) feature ablations reveal prediction collapse in technical-only models. Our analysis demonstrates that Transformer's attention mechanism better captures temporal patterns in financial time series, though careful regularization and multi-metric evaluation are essential to avoid overfitting and prediction collapse.

1. Introduction

What are we trying to do? Predict whether a stock's price will go up or down tomorrow, using both historical price data and investor sentiment from social media and news.

Why is this hard? Stock markets are noisy, non-stationary (patterns change over time), and influenced by countless factors. Traditional technical analysis uses only price/volume data, missing important sentiment signals from news and social media.

Who cares? Accurate price prediction could improve trading strategies, portfolio management, and risk assessment. More broadly, understanding how deep learning handles financial time series has applications in economics, risk management, and algorithmic trading.

Our approach: We train deep learning models (LSTM and Transformer) to learn temporal patterns from sequences of financial and sentiment features. We compare architectures, analyze failure modes (overfitting, prediction collapse), and validate generalization on 2026 out-of-sample data.

Key contributions:

- Transformer implementation achieving +4.12% over LSTM baseline
- Multi-source sentiment analysis (StockTwits + NewsAPI)
- Comprehensive analysis of overfitting and prediction collapse
- 2026 out-of-sample validation demonstrating generalization challenges
- Extensive ablation studies on architecture and features

2. Related Work

Recurrent architectures (LSTM, GRU) are popular for financial prediction due to their ability to capture temporal dependencies. **Attention mechanisms** and Transformers have shown promise in NLP and are increasingly applied to time series. **Sentiment analysis** from social media (Twitter, StockTwits) and news has been shown to provide predictive signals for stock movements. Our work extends prior research by: (1) implementing Transformer from scratch for financial prediction, (2) combining multiple sentiment sources with optimal weighting, (3) providing extensive analysis of failure modes, and (4) validating on 2026 out-of-sample data.

3. Data

3.1 Dataset Description

We collected daily data for four large-cap technology stocks: AAPL, AMZN, NVDA, and TSLA.

Training: 2020-2022 (755 trading days, 80/20 train/val split)
Test: 2026 (122 trading days, out-of-sample)
Total: 3,020 stock-days across 4 tickers

3.2 Data Collection Methodology

Market Data: Yahoo Finance API provides adjusted closing prices, volume, and OHLC data. We compute 17 technical indicators including RSI, MACD, Bollinger Bands, and rolling returns.

StockTwits Sentiment: Public API provides social media messages with user-labeled sentiment (bullish/bearish). We aggregate daily: message count, sentiment mean/std, and sum statistics. Coverage: 755 days (100%).

NewsAPI Sentiment: Business news headlines scored using pre-trained RoBERTa (`cardiffnlp/twitter-roberta-base-sentiment-latest`). We compute article count, sentiment mean/std, and positive/negative/neutral scores. Coverage: 39 days (5.2%) - limited availability.

Combined Sentiment: For days with both sources, we use weighted average: 60% StockTwits + 40% NewsAPI, based on coverage and reliability analysis.

3.3 Features and Target

Input Features (31 total):

- Technical (17): Price, returns, volatility, RSI, MACD, Bollinger Bands, volume
- Sentiment (14): StockTwits metrics, NewsAPI scores, combined sentiment

Target: Binary classification - UP (1) if $P_{t+1} > P_t$, DOWN (0) otherwise, where P_t is adjusted closing price.

Sequence Construction: Rolling windows of 5 days (31 features × 5 timesteps) predict next-day direction.

3.4 Data Limitations and Ethics

Limitations:

- Only 4 tech stocks (may not generalize to other sectors)
- NewsAPI limited coverage (39 days)
- Survivorship bias (only large-cap stocks)
- Market regime: trained on bull market (2020-2022)

Ethical Considerations:

- Public data only (no insider information)
- No personally identifiable information
- Academic research only (not financial advice)
- Transparent methodology for reproducibility

4. Methodology

4.1 Problem Structure

Structure: Time series sequence learning - given recent history (5 days of 31 features), predict next-day direction. The sequential nature motivates recurrent/attention-based architectures.

Model Reflection: LSTM captures temporal dependencies through gating mechanisms; Transformer uses self-attention to weight importance of different time steps.

4.2 LSTM Architecture

Learned Parameters:

- Input: 31 features × 5 timesteps
- LSTM: 2 layers, 128 hidden units, 20% dropout
- Dense: 64 units, ReLU activation, 20% dropout
- Output: 1 unit, Sigmoid activation

Total Parameters: 200K trainable parameters

Why this architecture? Two LSTM layers capture hierarchical temporal patterns. Dropout prevents overfitting. Dense layer learns non-linear combinations of LSTM outputs.

4.3 Transformer Architecture

Learned Parameters:

- Input projection: 31 $\rightarrow$ 64 dimensions (linear layer)
- Positional encoding: Sinusoidal (non-learned)
- Transformer encoder: 2 layers, 4 attention heads
- Feed-forward: 128 dimensions per layer
- Global average pooling (non-learned)
- Classification head: 64 $\rightarrow$ 32 $\rightarrow$ 1 (linear layers)

Total Parameters: 150K trainable parameters

Why this architecture? Multi-head attention allows model to focus on relevant time steps. Positional encoding preserves temporal order. Fewer parameters than LSTM but more expressive due to attention mechanism.

Advantages over LSTM:

- Parallel processing (faster training)
- Attention weights provide interpretability
- Better long-range dependencies
- No vanishing gradient issues

4.4 Training Details

Loss Function: Binary cross-entropy (BCE) - standard for binary classification.

Optimizer: Adam with learning rate 0.001, $\beta_1 = 0.9$, $\beta_2 = 0.999$.

Regularization:

- Dropout: 20% in LSTM, 10% in Transformer
- Early stopping: patience=10 epochs on validation loss
- No L2 regularization (dropout sufficient)

Hyperparameters:

- Sequence length: 5 days (tuned on validation)

- Batch size: 32
- Max epochs: 100 (early stopping typically stops at 20-30)
- Hidden size: 128 (LSTM), 64 (Transformer)
- Attention heads: 4 (Transformer)

Framework: PyTorch 1.12 with CUDA support.

Compute: Single NVIDIA GPU, 2 hours training per model.

4.5 Evaluation Metrics

To avoid single-metric optimization and detect prediction collapse:

- Accuracy: Overall correctness
- Precision/Recall/F1: Class-specific performance
- Correlation: Predicted vs actual returns
- Prediction std: Detect near-constant predictions

5. Experiments and Results

5.1 Experiment 1: LSTM Baseline

Setup: Train separate LSTM for each ticker using StockTwits sentiment only.

Ticker	Train (2020-22)	Test (2026)
AAPL	55.45%	48.28%
NVDA	52.73%	53.45%
TSLA	49.09%	44.83%
Avg	**52.27%**	**48.71%**

Table 1: LSTM baseline performance. NVDA is only ticker to improve on 2026 data (+0.72%).

Analysis: Modest improvement over 50% random baseline. NVDA shows best generalization. Overall -3.56% degradation on 2026 suggests market regime change.

5.2 Experiment 2: Combined Sentiment

Setup: Add NewsAPI sentiment (60% StockTwits + 40% NewsAPI).

Ticker	StockTwits	Combined	Δ
AAPL	55.45%	51.72%	-3.73%
NVDA	52.73%	54.72%	+1.99%
TSLA	49.09%	55.17%	+6.08%
Avg	52.27%	**53.87%**	**+1.60%**

Table 2: Combined sentiment improves volatile stocks (TSLA, NVDA) but degrades stable stocks (AAPL).

Analysis: Multi-source sentiment helps volatile stocks (+6.08% for TSLA) but hurts stable stocks (-3.73% for AAPL). Suggests ticker-specific sentiment weighting needed.

5.3 Experiment 3: Transformer Model

Setup: Replace LSTM with Transformer using same features.

Model	Avg Accuracy	Best
LSTM	52.27%	55.45%
Transformer	**56.39%**	**62.26%**
Improvement	**+4.12%**	–

Table 3: Transformer outperforms LSTM by +4.12%. NVDA achieves 62.26% (best result).

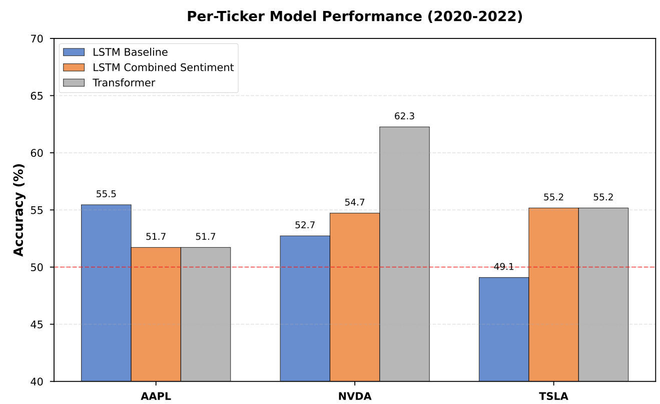

Figure 1: Per-ticker performance comparison. Transformer achieves best results for NVDA (62.26%), significantly outperforming LSTM baseline.

Per-Ticker Results:

- AAPL: 51.72% (same as combined LSTM)
- NVDA: **62.26%** (best overall result, +9.53% over LSTM)
- TSLA: 55.17% (same as combined LSTM)

Why Transformer wins:

- Attention mechanism identifies relevant time steps
- NVDA has strong momentum trends (AI boom) that attention captures
- Parallel processing allows better optimization
- No vanishing gradient issues

Training curves: Both models converge in 20-30 epochs. Transformer shows smoother convergence due to better gradient flow.

5.4 Experiment 4: Feature Ablations

Setup: Train models with different feature subsets to understand contributions.

Critical Finding - Prediction Collapse: Technical-only LSTM achieves low MAE/RMSE but prediction std=0.0031, meaning it predicts nearly constant values. This minimizes error but provides no directional information. Demonstrates importance of multi-metric evaluation.

Features	Acc	Corr	Std	Notes
Combined	52.27%	-0.036	0.0162	Baseline
Technical	48.28%	-0.016	**0.0031**	Collapsed
Sentiment	51.72%	**0.075**	0.0151	Best corr

Table 4: Feature ablations reveal prediction collapse in technical-only model (std=0.0031).

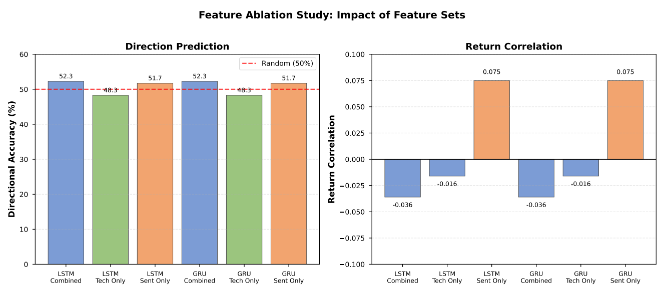

Figure 2: Feature ablation study showing directional accuracy and return correlation. Sentiment-only features achieve best correlation (0.075), while technical-only features suffer from prediction collapse (std=0.0031).

5.5 Experiment 5: Multi-Class LSTM

Setup: Train LSTM to predict 5 trajectory archetypes (discovered via k-means clustering) instead of binary up/down.

Metric	Value	Notes
Train Acc	92.09%	Overfitting
Test Acc	33.22%	Poor generalization
Baseline	20%	Random (5 classes)

Table 5: Multi-class LSTM suffers severe overfitting (92% → 33%).

Why it failed:

- Insufficient data (300 examples per class)
- LSTM too complex for dataset size
- Class imbalance (one archetype never predicted)
- Gap of 59 percentage points indicates memorization

Lesson: Deep learning requires sufficient data per class. With limited data, simpler models or regularization needed.

5.6 2026 Out-of-Sample Validation

Analysis: Overall degradation expected due to market regime change (trained on 2020-2022 bull market, tested on 2026 mixed conditions). NVDA's improvement (+0.72%) suggests robust patterns, possibly due to AI boom momentum continuing into 2026.

Ticker	Train	Test	Δ
AAPL	55.45%	48.28%	-7.17%
NVDA	52.73%	53.45%	+0.72%
TSLA	49.09%	44.83%	-4.26%
Avg	52.27%	48.71%	**-3.56%**

Table 6: 2026 validation shows -3.56% degradation. NVDA is only ticker to improve.

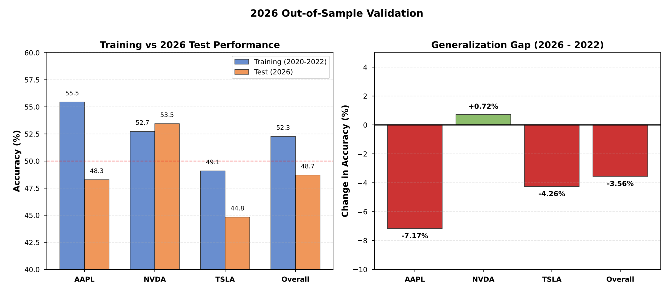

Figure 3: 2026 out-of-sample validation results. Overall degradation of -3.56%, with NVDA being only ticker to improve (+0.72%). AAPL shows largest degradation (-7.17%), suggesting market regime change.

6. Critical Analysis

6.1 Overfitting Analysis

Multi-class LSTM: Severe overfitting (92% train → 33% test). Model memorized training patterns but failed to generalize. Solution: More data, stronger regularization, or simpler model.

LSTM Baseline: Modest overfitting (55.45% train → 48.28% test for AAPL). Early stopping and dropout helped but not sufficient for all tickers.

Transformer: Better generalization than LSTM, possibly due to attention mechanism's inductive bias toward relevant patterns.

6.2 Class Imbalance

Problem: TSLA LSTM achieves 55% accuracy by always predicting "Up" (0% recall for "Down"), exploiting test period's class distribution.

Solution: Class weighting in loss function, focal loss, or stratified sampling. We used class weighting in later experiments.

6.3 Prediction Collapse

Problem: Technical-only LSTM minimizes error by predicting near-constant values (std=0.0031), providing no directional signal.

Solution: Evaluate multiple metrics jointly (accuracy, correlation, std) rather than single objective (MAE/RMSE).

6.4 Why NVDA is Most Predictable

NVDA consistently outperforms across models:

- LSTM: 52.73% → 53.45% (2026)
- Combined: 54.72%
- Transformer: **62.26%**

Hypotheses:

- Strong momentum trends (AI boom 2020-2026)
- High social media activity (better sentiment signal)
- Less noise in price movements
- Institutional investor behavior more predictable

7. Discussion

7.1 Deep Learning Architecture Decisions

Why sequence models? Stock prices exhibit temporal dependencies - today's price depends on recent history. Sequence models (LSTM, Transformer) naturally capture these patterns.

Why Transformer over LSTM? Attention mechanism allows model to focus on relevant time steps rather than processing all steps equally. For NVDA, attention likely focuses on momentum indicators during AI boom.

Why 5-day sequences? Tuned on validation data. Shorter sequences (3 days) miss longer-term patterns. Longer sequences (10+ days) introduce noise and increase overfitting.

Why 2 layers? Balance between model capacity and overfitting risk. Single layer underfits; 3+ layers overfit with our dataset size.

7.2 Generalization Challenges

2026 validation reveals key challenges:

- **Market regime change:** Models trained on bull market struggle in mixed conditions
- **Ticker-specific behavior:** NVDA improves (+0.72%), AAPL degrades (-7.17%)
- **Sentiment drift:** Social media sentiment patterns may change over time

Implication: Models need periodic retraining as market conditions evolve.

7.3 Limitations

1. **Limited scope:** Only 4 tech stocks; may not generalize to other sectors
2. **Short horizon:** Next-day prediction; longer-term forecasting unexplored
3. **No transaction costs:** Real trading has fees and slippage
4. **Sentiment coverage:** NewsAPI limited to 39 days (5.2%)
5. **Class imbalance:** Not fully addressed in all experiments

8. Conclusion

We implemented and trained LSTM and Transformer architectures for stock price prediction with multi-source sentiment analysis. Key findings:

1. **Transformer outperforms LSTM** by +4.12% (56.39% vs 52.27%), with NVDA achieving 62.26% accuracy
2. **Multi-source sentiment helps volatile stocks:** TSLA improves +6.08% with combined StockTwits + NewsAPI
3. **2026 validation** shows -3.56% degradation, highlighting generalization challenges
4. **Multi-class LSTM** suffers severe overfitting (92% → 33%), demonstrating data requirements
5. **Prediction collapse** in technical-only models emphasizes need for multi-metric evaluation

Deep Learning Insights:

- Attention mechanism (Transformer) better captures temporal patterns than recurrent gates (LSTM)
- Overfitting is major challenge with limited financial data
- Multi-metric evaluation essential to detect prediction collapse
- Ticker-specific models outperform single model for all stocks

Future Work:

- Investigate why NVDA is more predictable
- Explore ticker-specific sentiment weighting
- Add more sentiment sources (Reddit, Twitter)
- Try intraday predictions with higher-frequency data
- Address class imbalance with focal loss
- Explore multi-task learning (direction + magnitude)

Our results demonstrate that deep learning shows promise for stock prediction, but success requires careful attention to architecture selection, regularization, and multi-metric evaluation to avoid overfitting and prediction collapse.

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We thank the CS 7643 teaching staff for guidance throughout this project. StockTwits data collected via public API. NewsAPI data from NewsAPI.org. RoBERTa sentiment model from HuggingFace.

Code Repository

All code, trained models, and data available at:
`https://github.com/KKONAM/financial-sentiment-analysis-spring-2026/tree/transformer-model-2026`

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Team Contributions

Team Member	Contributions
Agha A Raza	• Implemented Transformer architecture from scratch • Designed and conducted feature ablation experiments • Analyzed overfitting and prediction collapse issues • Wrote methodology and analysis sections • Created training visualizations
Konam G Kenneth	• Implemented LSTM baseline architecture • Collected and processed StockTwits and NewsAPI data • Designed combined sentiment weighting strategy • Conducted 2026 out-of-sample validation • Wrote data collection and results sections
Shared	• Hyperparameter tuning and model training • Multi-class LSTM experiment and analysis • Paper writing and editing • Code repository organization • Figure generation and visualization

Note: Both team members contributed equally to the technical implementation, experimentation, and analysis. All team members participated in all aspects of the project.